FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	79.69	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	79.69	0.00	0.00000

==========

COLUMN NO. 5 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	134.98	55.65	0.13301

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	134.98	1.77	0.00424

==========

COLUMN NO. 6 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	79.69	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	79.69	0.00	0.00000

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COLUMN NO. 7 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	110.01	2.68	0.00784

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	110.01	1.22	0.00359

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COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.077 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	79.69	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	79.69	0.00	0.00000

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COLUMN NO. 9 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	353.80	43.17	0.03874

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	353.80	7.20	0.00646

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COLUMN NO. 10 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	338.65	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	338.65	0.00	0.00000

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COLUMN NO. 11 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	348.78	34.01	0.03001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	348.78	15.59	0.01376

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COLUMN NO. 12 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	285.26	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	285.26	0.00	0.00000

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COLUMN NO. 13 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 4053.6 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.340	304	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12593.17	10074.53	4523.92	1106.53	244.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
518.62	-2026.73	207.03	72.65	0.12997

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

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   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12593.17  10074.53   4523.92   1106.53     244.6
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  518.62  -2026.73    207.03    451.52    0.80776
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COLUMN NO. 14 DESIGN PER ACI 318-08 - AXIAL + BENDING

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FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 12 - 20 MM               1.047      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    129.82      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
12181.21   9744.97   4499.60   1041.49     231.5
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  411.95  -1583.37    129.82      0.00    0.00000
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COLUMN NO. 15 DESIGN PER ACI 318-08 - AXIAL + BENDING

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FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 600.0 X 600.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3600.0  SQ. MM
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BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	610.75	194.74	0.11809

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	610.75	50.56	0.03066

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COLUMN NO. 16 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	185.60	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	185.60	0.00	0.00000

==

COLUMN NO. 17 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	369.85	30.78	0.02561

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	369.85	18.56	0.01544

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COLUMN NO. 18 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	334.26	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	334.26	0.00	0.00000

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COLUMN NO. 19 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	337.19	39.45	0.03600

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	337.19	3.78	0.00345

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COLUMN NO. 20 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 550.0 X 550.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
16 - 16 MM	1.063	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	289.86	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
10254.82	8203.86	3752.05	797.52	212.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
320.56	-1351.15	289.86	0.00	0.00000

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COLUMN NO. 21 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
AREA OF STEEL REQUIRED = 4104.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
24 - 16 MM	1.340	305	STA	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12593.17	10074.53	4523.92	1106.53	244.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
518.62	-2026.73	322.87	6.82	0.00782

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12593.17	10074.53	4523.92	1106.53	244.6
M0	P-tens.	Des.Pn	Des.Mn	e/h
518.62	-2026.73	322.87	521.19	0.59786

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COLUMN NO. 22 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	139.60	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	139.60	0.00	0.00000

==========

COLUMN NO. 23 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	450.76	88.84	0.07299

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	450.76	17.06	0.01401

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COLUMN NO. 24 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	192.80	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	192.80	0.00	0.00000

==

COLUMN NO. 25 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	155.28	36.16	0.07512

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	155.28	33.67	0.06996

==

COLUMN NO. 26 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	66.77	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	66.77	0.00	0.00000

==========

COLUMN NO. 27 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	115.75	31.44	0.08763

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	115.75	3.14	0.00875

==========

COLUMN NO. 28 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	75.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	75.38	0.00	0.00000

==========

COLUMN NO. 29 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	107.15	37.97	0.11431

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

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 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    107.15      8.89    0.02675
 ------------------------------------------------------
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COLUMN NO. 30 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
------------------------------------------------------------

 12 - 20 MM               1.077      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 320.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4455.00   1204.46     270.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  485.93  -1583.37    127.08      0.00    0.00000
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
11883.71   9506.97   4260.51    834.62     195.9
    M0       P-tens.   Des.Pn    Des.Mn     e/h
  337.39  -1583.37    127.08      0.00    0.00000
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COLUMN NO. 31 DESIGN PER ACI 318-08 - AXIAL + BENDING

```
FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 500.0 X 700.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  3500.0  SQ. MM
```

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	125.44	39.07	0.10046

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	125.44	7.52	0.01935

==========

COLUMN NO. 32 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 500.0 X 700.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3500.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.077	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4455.00	1204.46	270.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
485.93	-1583.37	75.38	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
11883.71	9506.97	4260.51	834.62	195.9
M0	P-tens.	Des.Pn	Des.Mn	e/h
337.39	-1583.37	75.38	0.00	0.00000

==========

COLUMN NO. 36 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	56.00	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	56.00	0.00	0.00000

==========

COLUMN NO. 37 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
48 - 16 MM	1.072	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

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 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
30541.33  24433.07  11905.66   4477.72     376.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1820.29  -4053.45    993.76    227.83    0.08491
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
30541.33  24433.07  11811.11   4009.21     339.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1626.70  -4053.45    993.76     29.94    0.01116
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COLUMN NO. 38 DESIGN PER ACI 318-08 - AXIAL + BENDING

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FY - 420.0  FC -  35.0 MPA,  RECT SIZE - 900.0 X1000.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  9000.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ---------------------------------------------------------

 48 - 16 MM               1.072      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
30541.33  24433.07  11905.66   4477.72     376.1
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1820.29  -4053.45   1367.69      0.00    0.00000
 ---------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ---------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
30541.33  24433.07  11811.11   4009.21     339.4
    M0       P-tens.   Des.Pn    Des.Mn     e/h
 1626.70  -4053.45   1367.69      0.00    0.00000
 ---------------------------------------------------------
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COLUMN NO. 39 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
48 - 16 MM	1.072	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11905.66	4477.72	376.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
1820.29	-4053.45	1081.81	225.09	0.07706

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11811.11	4009.21	339.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
1626.70	-4053.45	1081.81	9.03	0.00309

==========

COLUMN NO. 40 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
48 - 16 MM	1.072	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11905.66	4477.72	376.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
1820.29	-4053.45	1406.46	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11811.11	4009.21	339.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
1626.70	-4053.45	1406.46	0.00	0.00000

==========

COLUMN NO. 41 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
48 - 16 MM	1.072	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11905.66	4477.72	376.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
1820.29	-4053.45	1193.61	136.75	0.04243

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11811.11	4009.21	339.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
1626.70	-4053.45	1193.61	25.38	0.00787

==========

COLUMN NO. 42 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

48 - 16 MM 1.072 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11905.66	4477.72	376.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
1820.29	-4053.45	1369.85	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11811.11	4009.21	339.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
1626.70	-4053.45	1369.85	0.00	0.00000

==========

COLUMN NO. 43 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

48 - 16 MM 1.072 301 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11905.66	4477.72	376.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
1820.29	-4053.45	1307.24	107.57	0.03048

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11811.11	4009.21	339.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
1626.70	-4053.45	1307.24	18.40	0.00521

==========

COLUMN NO. 44 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, RECT SIZE - 900.0 X1000.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 9000.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
48 - 16 MM	1.072	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11905.66	4477.72	376.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
1820.29	-4053.45	1408.62	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
30541.33	24433.07	11811.11	4009.21	339.4
M0	P-tens.	Des.Pn	Des.Mn	e/h
1626.70	-4053.45	1408.62	0.00	0.00000

==========

COLUMN NO. 45 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	93.16	24.88	0.08615

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	93.16	2.63	0.00909

==========

COLUMN NO. 46 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	105.32	2.28	0.01546

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	105.32	9.94	0.06739

==========

COLUMN NO. 47 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	96.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	96.92	0.00	0.00000

==========

COLUMN NO. 48 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	107.01	1.82	0.01214

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	107.01	10.42	0.06954

==========

COLUMN NO. 49 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	99.08	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	99.08	0.00	0.00000

==========

COLUMN NO. 50 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	62.34	0.65	0.01163

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	62.34	0.68	0.01218

==========

COLUMN NO. 51 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	56.00	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	56.00	0.00	0.00000

==========

COLUMN NO. 52 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	61.42	0.19	0.00346

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	61.42	0.68	0.01237

==========

COLUMN NO. 53 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

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 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36     56.00      0.00    0.00000
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36     56.00      0.00    0.00000
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COLUMN NO. 54 DESIGN PER ACI 318-08 - AXIAL + BENDING

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FY - 420.0  FC -  35.0 MPA,  SQRE SIZE - 450.0 X 450.0 MMS, TIED
        ONLY MINIMUM STEEL IS REQUIRED.
        AREA OF STEEL REQUIRED =  2025.0  SQ. MM

BAR CONFIGURATION       REINF PCT.   LOAD   LOCATION   PHI
 ------------------------------------------------------

 12 - 16 MM               1.191      301      END     0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER   10 SPACING 256.00 MM

    COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36     19.31      0.22    0.00951
 ------------------------------------------------------

    COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

 ------------------------------------------------------
   P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
 6965.96   5572.77   2433.31    442.03     181.7
   M0        P-tens.   Des.Pn    Des.Mn     e/h
  191.29  -1013.36     19.31      1.42    0.06110
 ------------------------------------------------------
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COLUMN NO. 55 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	12.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	12.92	0.00	0.00000

==========

COLUMN NO. 56 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	28.51	1.99	0.05819

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	28.51	2.23	0.06519

==========

COLUMN NO. 57 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	12.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	12.92	0.00	0.00000

==========

COLUMN NO. 58 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	29.76	1.85	0.05188

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	29.76	2.48	0.06939

==========

COLUMN NO. 59 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	12.92	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	12.92	0.00	0.00000

==========

COLUMN NO. 60 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	20.47	0.01	0.00051

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
6965.96	5572.77	2433.31	442.03	181.7
M0	P-tens.	Des.Pn	Des.Mn	e/h
191.29	-1013.36	20.47	0.15	0.00592

==========

COLUMN NO. 61 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 450.0 X 450.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 2025.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 16 MM	1.191	301	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 256.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

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-------------------------------------------------------
  P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
6965.96   5572.77   2433.31    442.03     181.7
  M0        P-tens.   Des.Pn    Des.Mn     e/h
 191.29  -1013.36     12.92      0.00    0.00000
-------------------------------------------------------

   COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

-------------------------------------------------------
  P0        Pn max    P-bal.    M-bal.     e-bal. (MM)
6965.96   5572.77   2433.31    442.03     181.7
  M0        P-tens.   Des.Pn    Des.Mn     e/h
 191.29  -1013.36     12.92      0.00    0.00000
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********************END OF COLUMN DESIGN RESULTS********************


608. DESIGN BEAM 71 73 TO 83 86 TO 102
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BEAM NO. 71 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

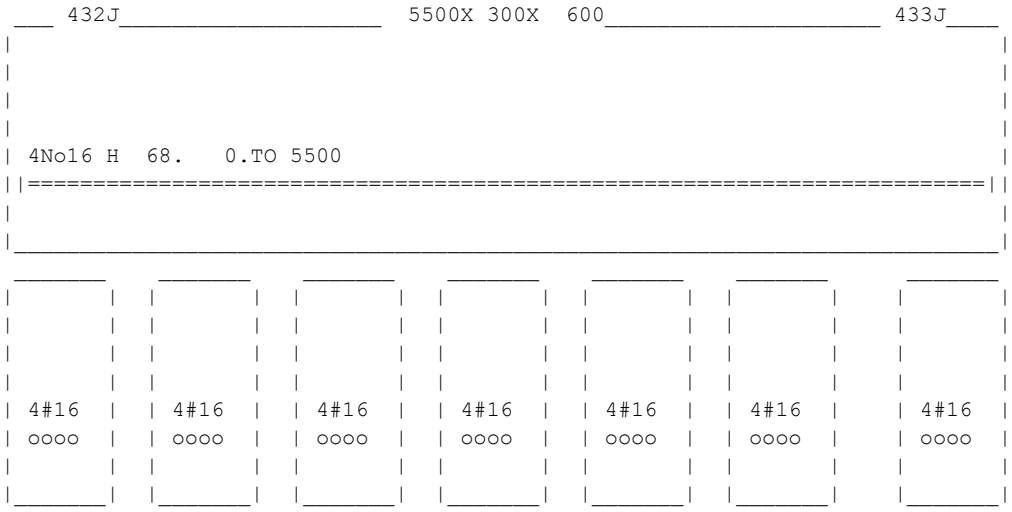
LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	4 - 16MM	0.	5500.	YES	YES

CRITICAL POS MOMENT= 121.01 KN-MET AT 5500.MM, LOAD 303
REQD STEEL= 619.MM2, RHO=0.0039, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 838. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX NEG MOMENT = 160.22 KN-MET, LOADING 303



BEAM NO. 73 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1469. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	54.	2 - 12MM	0.	1469.	YES	YES

CRITICAL POS MOMENT=	24.11 KN-MET AT 0.MM, LOAD 304
REQD STEEL=	156.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	255./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH =	308. MMS

Cracked Moment of Inertia Iz at above location = 40000.2 cm^4

2	546.	2 - 12MM	0.	1469.	YES	YES

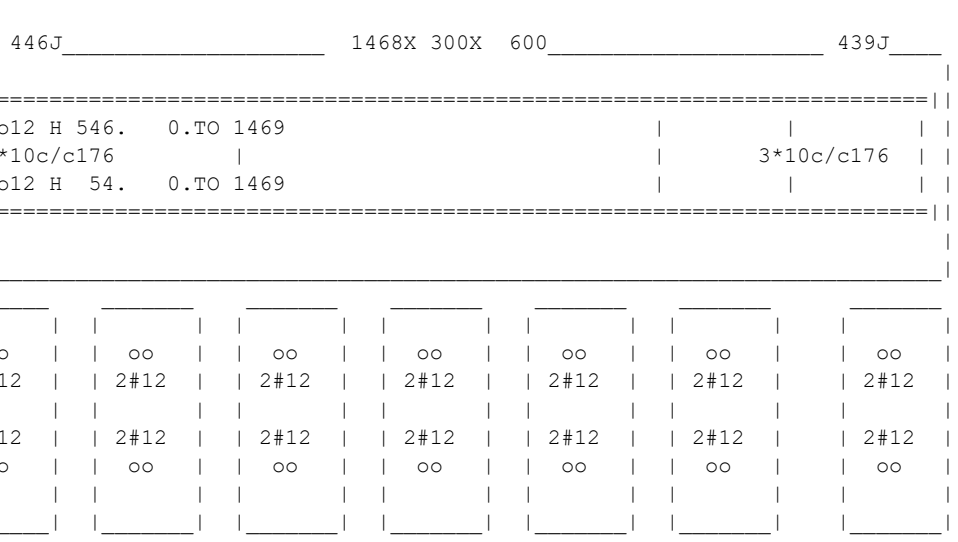
CRITICAL NEG MOMENT=	29.09 KN-MET AT 1469.MM, LOAD 303
REQD STEEL=	189.MM2, RHO=0.0012, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	255./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH =	308. MMS

Cracked Moment of Inertia Iz at above location = 40000.2 cm^4

B E A M N O. 73 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.03 KNS Vc= 160.79 KNS Vs= 0.00 KNS
Tu= 10.80 KN-MET Tc= 6.6 KN-MET Ts= 14.4 KN-MET LOAD 305
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 176. MM C/C FOR 193. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.91 SQ.CM.

AT END SUPPORT - Vu= 4.07 KNS Vc= 160.79 KNS Vs= 0.00 KNS
Tu= 10.80 KN-MET Tc= 6.6 KN-MET Ts= 14.4 KN-MET LOAD 305
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 176. MM C/C FOR 193. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.91 SQ.CM.



BEAM NO. 74 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	5500.	YES	YES
	CRITICAL POS MOMENT= 104.02 KN-MET AT 5500.MM, LOAD 303					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 85862.6 cm^4					
2	532.	4 - 16MM	0.	2825.	YES	NO
	CRITICAL NEG MOMENT= 135.88 KN-MET AT 0.MM, LOAD 303					
	REQD STEEL= 697.MM2, RHO=0.0044, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS					
	REQD. DEVELOPMENT LENGTH = 601. MMS					
	Cracked Moment of Inertia Iz at above location = 114685.2 cm^4					
3	534.	5 - 12MM	2579.	5500.	NO	YES

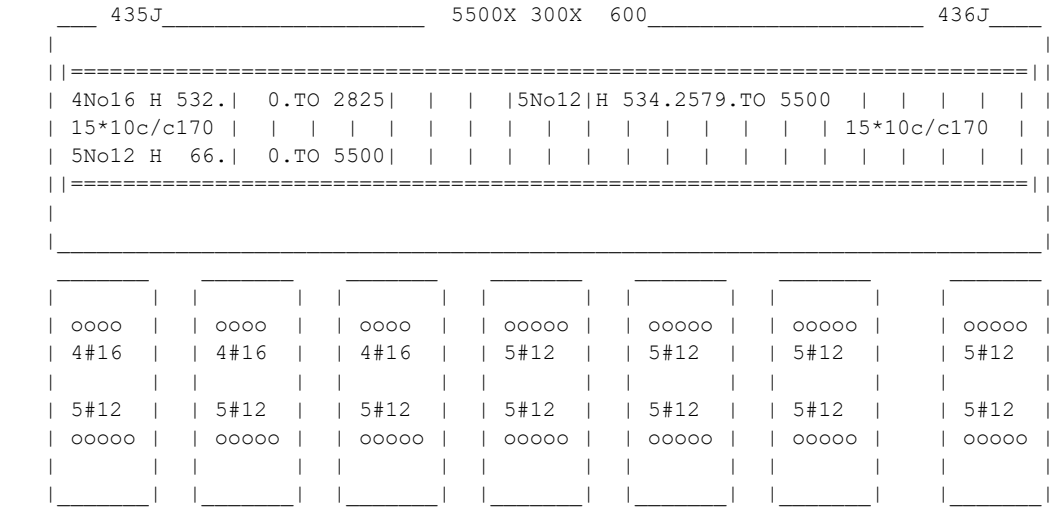
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|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    101.08 KN-MET  AT  5500.MM, LOAD  306|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS        |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                       |
|-------------------------------------------------------------|

 Cracked Moment of Inertia Iz at above location =    85862.6 cm^4
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B E A M N O. 74 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 23.80 KNS Vc= 146.37 KNS Vs= 0.00 KNS
Tu= 7.46 KN-MET Tc= 6.2 KN-MET Ts= 9.9 KN-MET LOAD 305
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 2220. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.04 SQ.CM.

AT END SUPPORT - Vu= 1.21 KNS Vc= 146.37 KNS Vs= 0.00 KNS
Tu= 7.46 KN-MET Tc= 6.2 KN-MET Ts= 9.9 KN-MET LOAD 305
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 170. MM C/C FOR 2220. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 2.04 SQ.CM.



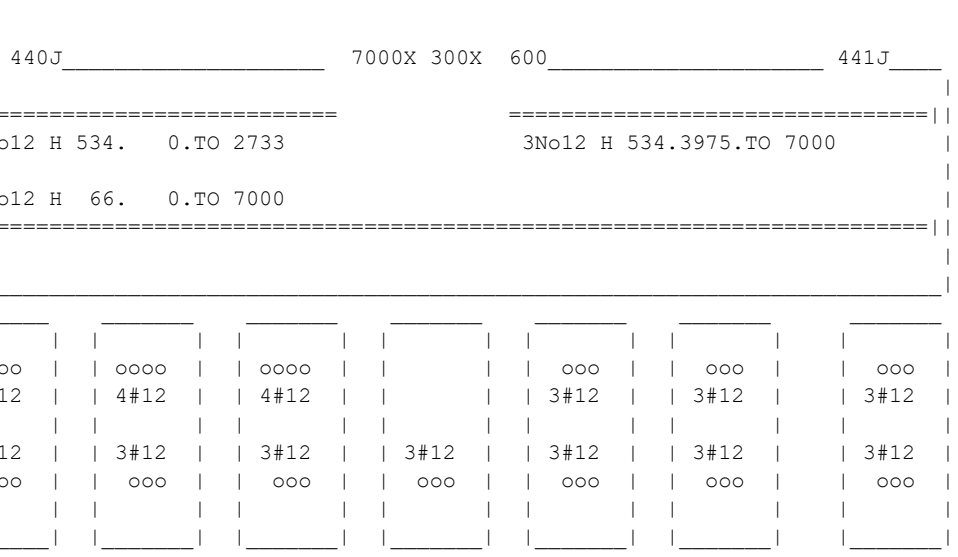
BEAM NO. 75 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	7000.	YES	YES
	CRITICAL POS MOMENT= 39.40 KN-MET AT 6417.MM, LOAD 303					
	REQD STEEL= 262.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 54977.0 cm^4					
2	534.	4 - 12MM	0.	2733.	YES	NO
	CRITICAL NEG MOMENT= 63.24 KN-MET AT 0.MM, LOAD 303					
	REQD STEEL= 423.MM2, RHO=0.0026, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 70807.2 cm^4					
3	534.	3 - 12MM	3975.	7000.	NO	YES
	CRITICAL NEG MOMENT= 43.71 KN-MET AT 7000.MM, LOAD 302					
	REQD STEEL= 291.MM2, RHO=0.0018, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 54977.0 cm^4					

B E A M N O. 75 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   21.26 KNS  Vc=  148.69 KNS  Vs=    0.00 KNS
Tu=    1.62 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   14.01 KNS  Vc=  148.69 KNS  Vs=    0.00 KNS
Tu=    1.62 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                    STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 76 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	2 - 16MM	0.	1500.	YES	YES

CRITICAL POS MOMENT= 35.32 KN-MET AT 0.MM, LOAD 308
REQD STEEL= 349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 23144.3 cm^4

2	332.	2 - 16MM	0.	1500.	YES	YES

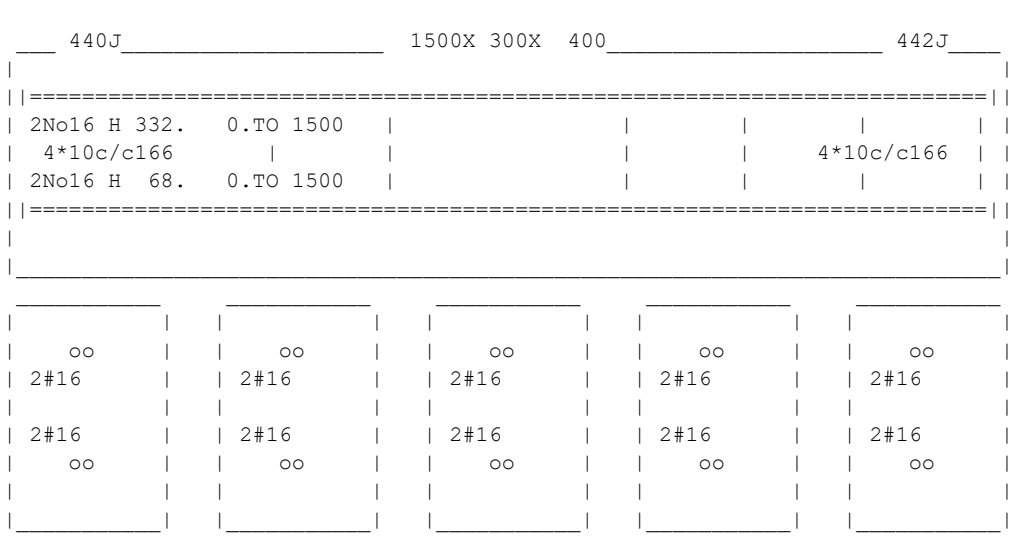
CRITICAL NEG MOMENT= 38.67 KN-MET AT 1500.MM, LOAD 304
REQD STEEL= 349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 23144.3 cm^4

B E A M N O. 76 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 47.87 KNS Vc= 97.98 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 304
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 166. MM C/C FOR 420. MM

AT END SUPPORT - Vu= 50.72 KNS Vc= 97.75 KNS Vs= 0.00 KNS
Tu= 0.08 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 304
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 166. MM C/C FOR 420. MM



BEAM NO. 77 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	2 - 16MM	0.	1500.	YES	YES

CRITICAL POS MOMENT= 38.87 KN-MET AT 1500.MM, LOAD 305
REQD STEEL= 349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 23144.3 cm^4

2	332.	2 - 16MM	0.	1500.	YES	YES

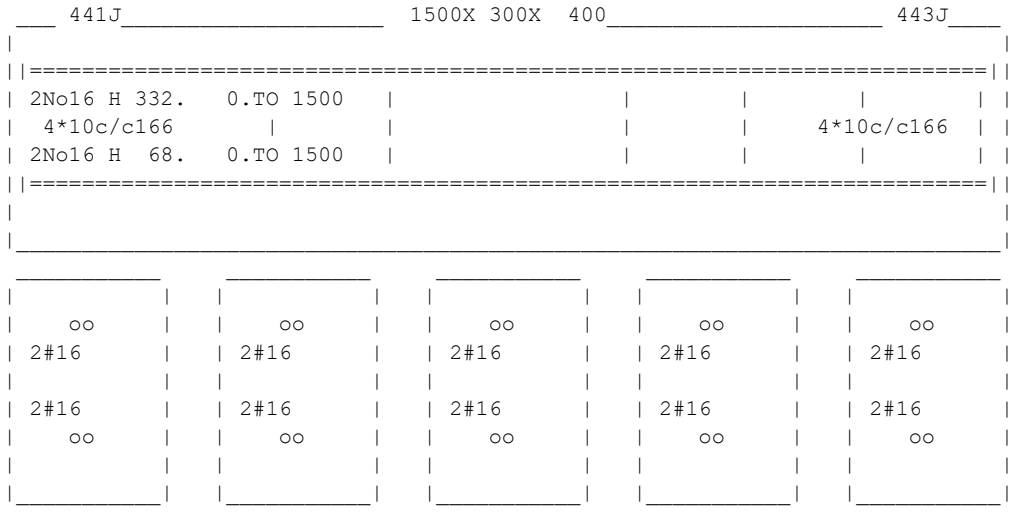
CRITICAL NEG MOMENT= 38.23 KN-MET AT 0.MM, LOAD 305
REQD STEEL= 349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 23144.3 cm^4

B E A M N O. 77 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 44.19 KNS Vc= 97.84 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 304
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 166. MM C/C FOR 420. MM

AT END SUPPORT - Vu= 47.04 KNS Vc= 97.80 KNS Vs= 0.00 KNS
Tu= 0.23 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 304
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 166. MM C/C FOR 420. MM



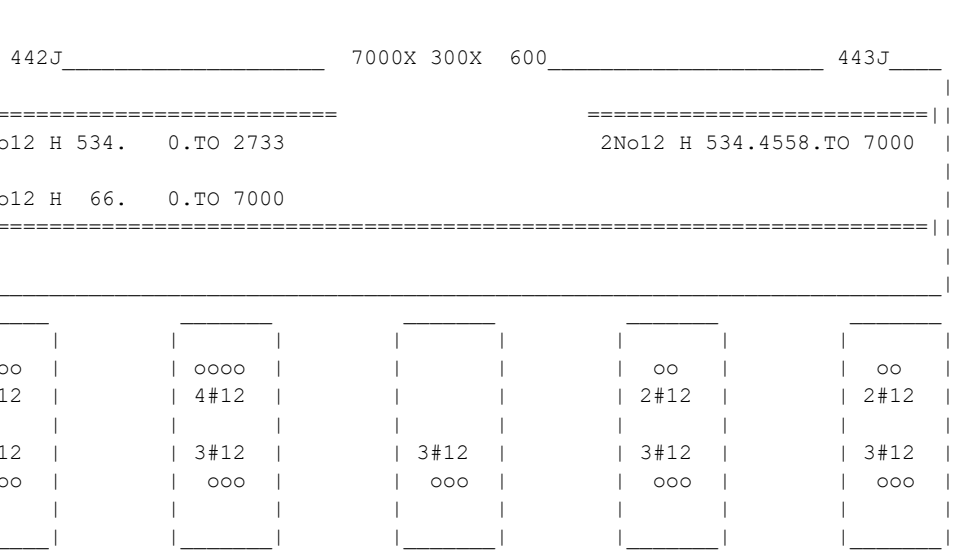
BEAM NO. 78 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	7000.	YES	YES
	CRITICAL POS MOMENT= 37.09 KN-MET AT 6417.MM, LOAD 303					
	REQD STEEL= 246.MM2, RHO=0.0015, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 54977.0 cm^4					
2	534.	4 - 12MM	0.	2733.	YES	NO
	CRITICAL NEG MOMENT= 62.02 KN-MET AT 0.MM, LOAD 303					
	REQD STEEL= 414.MM2, RHO=0.0026, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 70807.2 cm^4					
3	534.	2 - 12MM	4558.	7000.	NO	YES
	CRITICAL NEG MOMENT= 32.76 KN-MET AT 7000.MM, LOAD 306					
	REQD STEEL= 217.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 38198.0 cm^4					

B E A M N O. 78 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   23.08 KNS  Vc=  150.06 KNS  Vs=    0.00 KNS
Tu=    1.44 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   12.19 KNS  Vc=  150.06 KNS  Vs=    0.00 KNS
Tu=    1.44 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                    STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 79 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

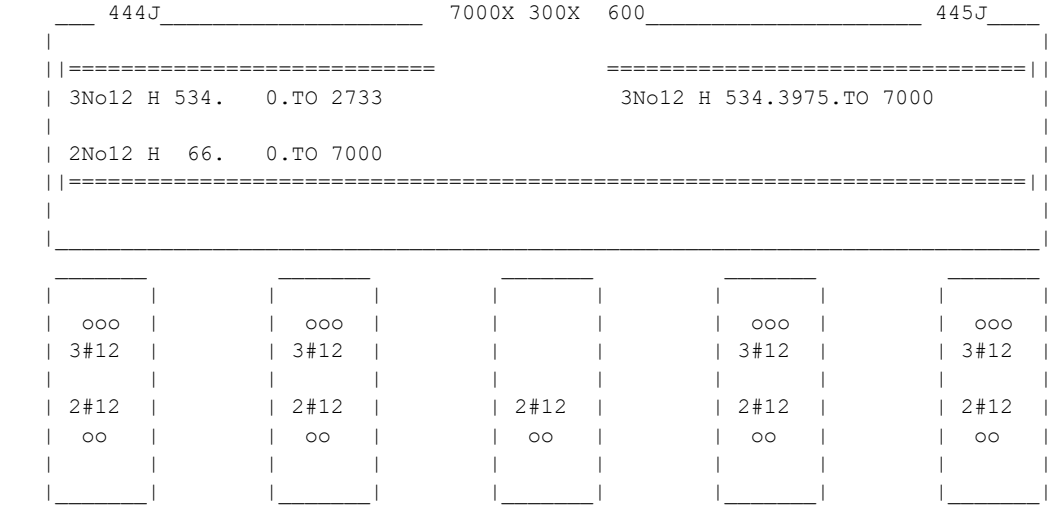
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	7000.	YES	YES
	CRITICAL POS MOMENT= 17.69 KN-MET AT 4667.MM, LOAD 303					
	REQD STEEL= 117.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 38198.0 cm^4					
2	534.	3 - 12MM	0.	2733.	YES	NO
	CRITICAL NEG MOMENT= 41.30 KN-MET AT 0.MM, LOAD 303					
	REQD STEEL= 275.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 54977.0 cm^4					
3	534.	3 - 12MM	3975.	7000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     41.36 KN-MET  AT  7000.MM, LOAD  302|
|   REQD STEEL=   275.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   84. MMS        |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 54977.0 cm^4

B E A M N O. 79 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 18.04 KNS Vc= 151.40 KNS Vs= 0.00 KNS
Tu= 0.55 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 17.23 KNS Vc= 151.85 KNS Vs= 0.00 KNS
Tu= 0.55 KN-MET Tc= 6.7 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 80 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	579.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=      5.46 KN-MET  AT     0.MM, LOAD  308|
|   REQD STEEL=    58.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14234.5 cm^4

  2         68.     2 - 16MM       331.         1500.        NO   YES
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=     42.14 KN-MET  AT  1500.MM, LOAD  305|
|   REQD STEEL=   349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./  164. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    23144.3 cm^4

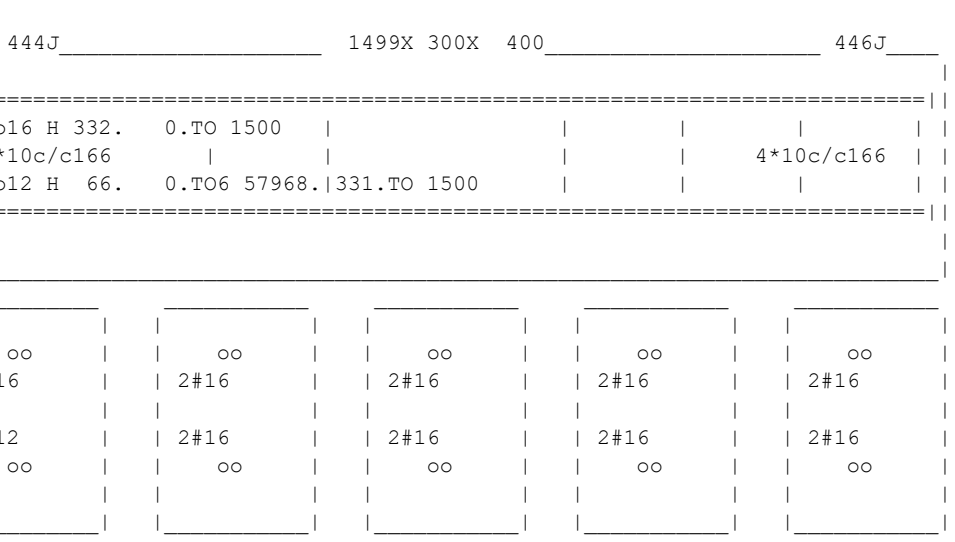
  3        332.     2 - 16MM         0.         1500.        YES  YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     39.47 KN-MET  AT     0.MM, LOAD  305|
|   REQD STEEL=   349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./  164. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    23144.3 cm^4
```

B E A M N O. 80 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   38.11 KNS  Vc=   99.11 KNS  Vs=    0.00 KNS
Tu=    2.41 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD   302
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 166. MM  C/C FOR  420. MM

AT END   SUPPORT - Vu=   52.98 KNS  Vc=   98.11 KNS  Vs=    0.00 KNS
Tu=    0.86 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD   305
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 166. MM  C/C FOR  420. MM
```



BEAM NO. 81 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	902.	YES	NO
	CRITICAL POS MOMENT= 28.50 KN-MET AT 0.MM, LOAD 308					
	REQD STEEL= 305.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 20273.6 cm^4					
2	66.	3 - 12MM	542.	1500.	NO	YES
	CRITICAL POS MOMENT= 27.90 KN-MET AT 1500.MM, LOAD 309					
	REQD STEEL= 299.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 20273.6 cm^4					
3	332.	2 - 16MM	0.	1500.	YES	YES

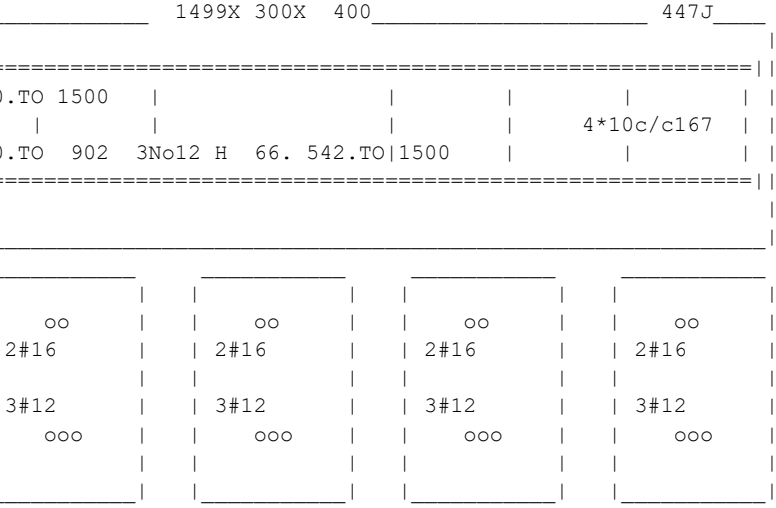
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|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     35.07 KN-MET  AT  1500.MM, LOAD  304|
|   REQD STEEL=   349.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./  164. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    23144.3 cm^4
```

B E A M N O. 81 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   40.73 KNS  Vc=   98.74 KNS  Vs=    0.00 KNS
Tu=    0.35 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   304
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM  C/C FOR  420. MM

AT END   SUPPORT - Vu=   43.58 KNS  Vc=   97.46 KNS  Vs=    0.00 KNS
Tu=    0.35 KN-MET  Tc=    3.9 KN-MET  Ts=    0.0 KN-MET  LOAD   304
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 167. MM  C/C FOR  420. MM
```



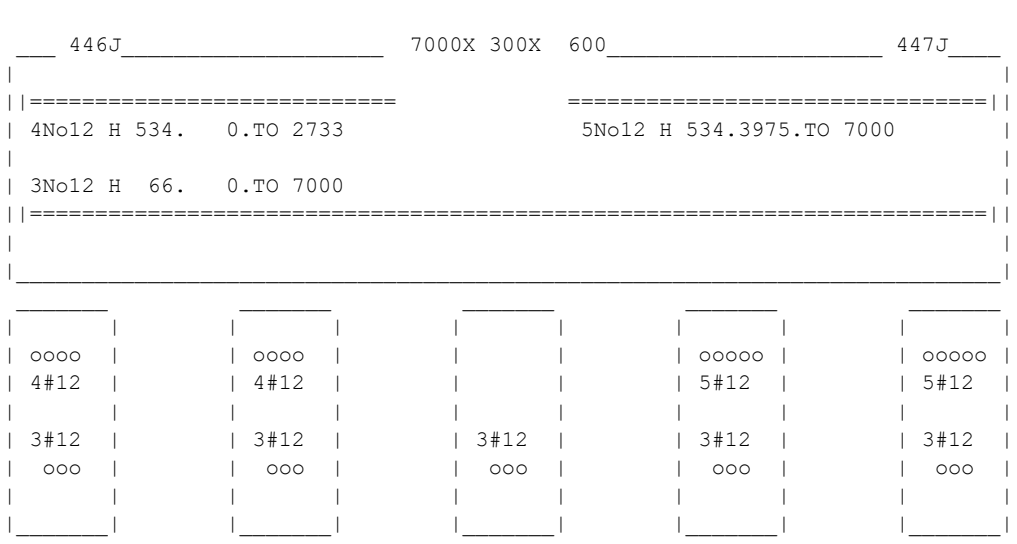
BEAM NO. 82 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 7000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	3 - 12MM	0.	7000.	YES	YES
	CRITICAL POS MOMENT= 38.58 KN-MET AT 0.MM, LOAD 306					
	REQD STEEL= 256.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 84. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 54977.0 cm^4					
2	534.	4 - 12MM	0.	2733.	YES	NO
	CRITICAL NEG MOMENT= 63.15 KN-MET AT 0.MM, LOAD 303					
	REQD STEEL= 422.MM2, RHO=0.0026, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 70807.2 cm^4					
3	534.	5 - 12MM	3975.	7000.	NO	YES
	CRITICAL NEG MOMENT= 70.70 KN-MET AT 7000.MM, LOAD 302					
	REQD STEEL= 473.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 85862.6 cm^4					

B E A M N O. 82 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   16.34 KNS  Vc=  157.83 KNS  Vs=    0.00 KNS
Tu=    0.11 KN-MET  Tc=    6.7 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   18.93 KNS  Vc=  153.15 KNS  Vs=    0.00 KNS
Tu=    0.11 KN-MET  Tc=    6.7 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                    STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 83 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

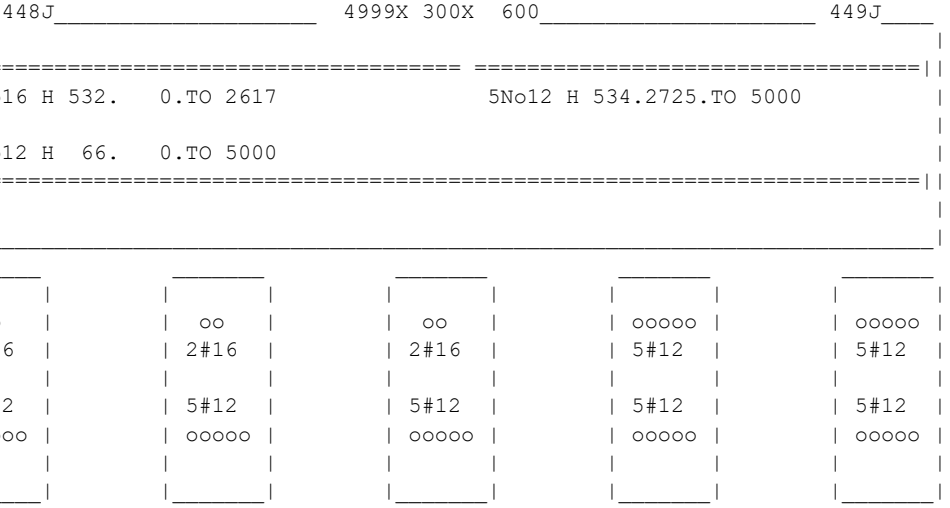
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	5000.	YES	YES
	CRITICAL POS MOMENT= 98.34 KN-MET AT 0.MM, LOAD 302					
	REQD STEEL= 562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS					
	REQD. DEVELOPMENT LENGTH = 440. MMS					
	Cracked Moment of Inertia Iz at above location = 85862.6 cm^4					
2	532.	2 - 16MM	0.	2617.	YES	NO
	CRITICAL NEG MOMENT= 60.05 KN-MET AT 0.MM, LOAD 307					
	REQD STEEL= 401.MM2, RHO=0.0025, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 63898.9 cm^4					
3	534.	5 - 12MM	2725.	5000.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     88.11 KN-MET  AT  5000.MM, LOAD  302|
|   REQD STEEL=   562.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    85862.6 cm^4
```

B E A M N O. 83 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    1.28 KNS  Vc=  148.56 KNS  Vs=    0.00 KNS
 Tu=    1.53 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   22.11 KNS  Vc=  148.56 KNS  Vs=    0.00 KNS
 Tu=    1.53 KN-MET  Tc=    6.3 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 86 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	5000.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     72.55 KN-MET  AT     0.MM, LOAD  302|
|   REQD STEEL=   486.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    85862.6 cm^4

  2        532.     2 - 16MM         0.         2617.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     51.68 KN-MET  AT     0.MM, LOAD  307|
|   REQD STEEL=   346.MM2, RHO=0.0022, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./  164. MMS         |
|   REQD. DEVELOPMENT LENGTH =   410. MMS                        |
|--------------------------------------------------------------|

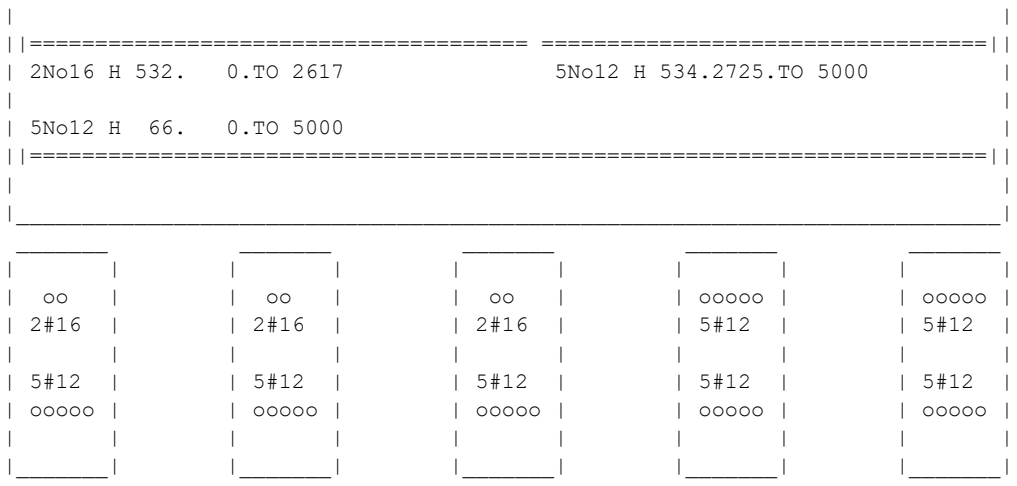
  Cracked Moment of Inertia Iz at above location =    63391.8 cm^4

  3        534.     5 - 12MM      2725.         5000.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     72.95 KN-MET  AT  5000.MM, LOAD  302|
|   REQD STEEL=   489.MM2, RHO=0.0030, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    85862.6 cm^4
```

B E A M N O. 86 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    2.58 KNS  Vc=  152.07 KNS  Vs=    0.00 KNS
Tu=    0.22 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   20.82 KNS  Vc=  152.07 KNS  Vs=    0.00 KNS
Tu=    0.22 KN-MET  Tc=    6.4 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



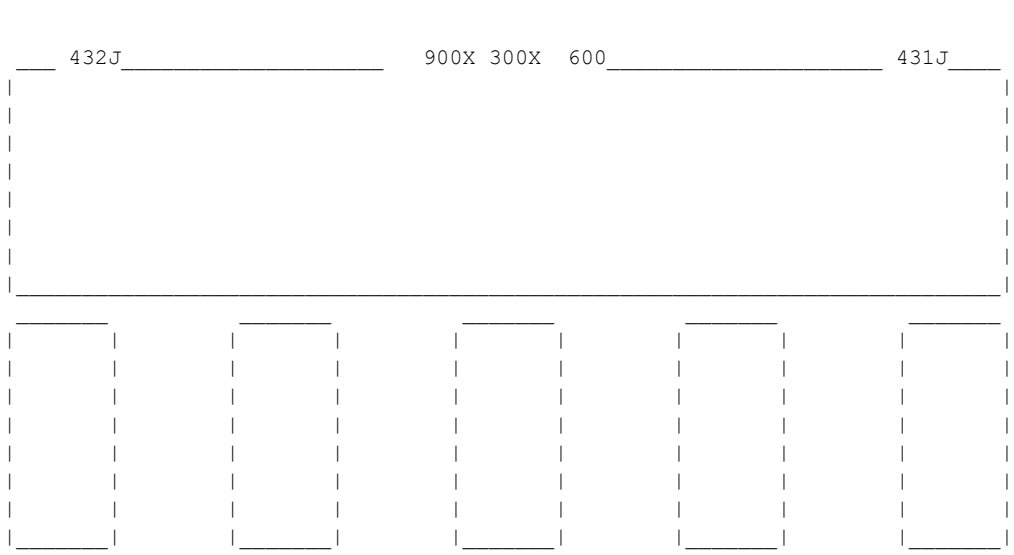
BEAM NO. 87 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 900. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 1214. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX POS MOMENT = 228.18 KN-MET, LOADING 302

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 1061. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX NEG MOMENT = 200.81 KN-MET, LOADING 305



BEAM NO. 88 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

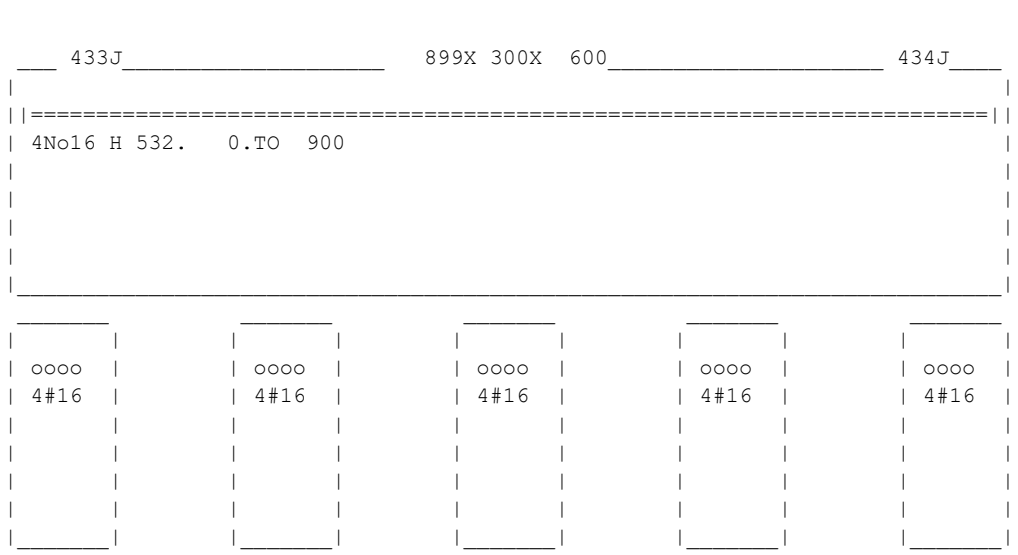
LEN - 900. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	532.	4 - 16MM	0.	900.	YES	YES

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 1422. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX POS MOMENT = 264.53 KN-MET, LOADING 303

CRITICAL NEG MOMENT=	140.46 KN-MET AT 0.MM, LOAD 305
REQD STEEL=	722.MM2, RHO=0.0045, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING=	225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH =	601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4



BEAM NO. 89 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 934. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	70.	YES	NO

CRITICAL POS MOMENT= 7.78 KN-MET AT 0.MM, LOAD 306
REQD STEEL= 51.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

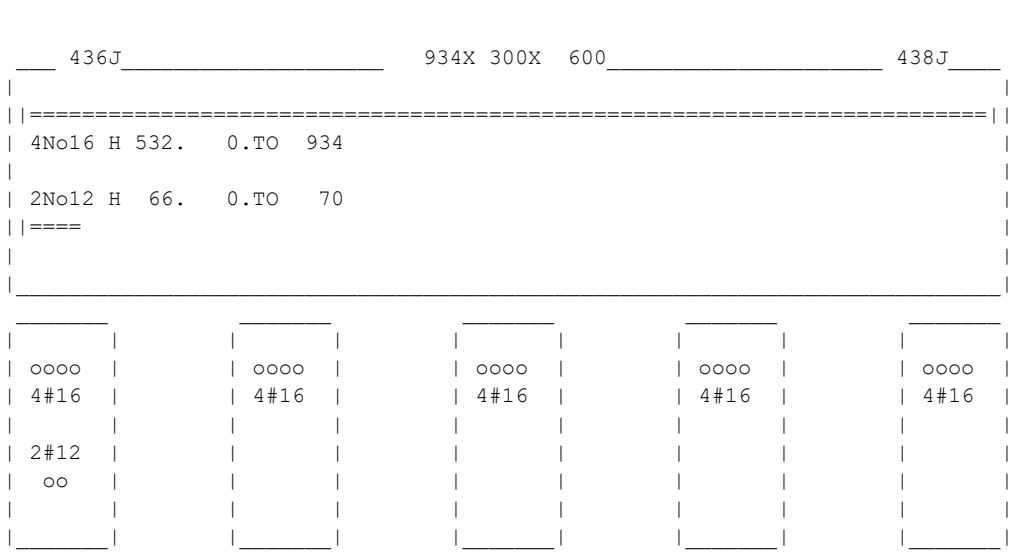
Cracked Moment of Inertia Iz at above location = 38198.0 cm^4

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 1136. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX POS MOMENT = 214.34 KN-MET, LOADING 303

2	532.	4 - 16MM	0.	934.	YES	YES

CRITICAL NEG MOMENT= 119.64 KN-MET AT 934.MM, LOAD 306
REQD STEEL= 611.MM2, RHO=0.0038, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 114685.2 cm^4



BEAM NO. 90 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 934. MM FY - 420. FC - 35. MPA, SIZE - 300. X 600. MMS

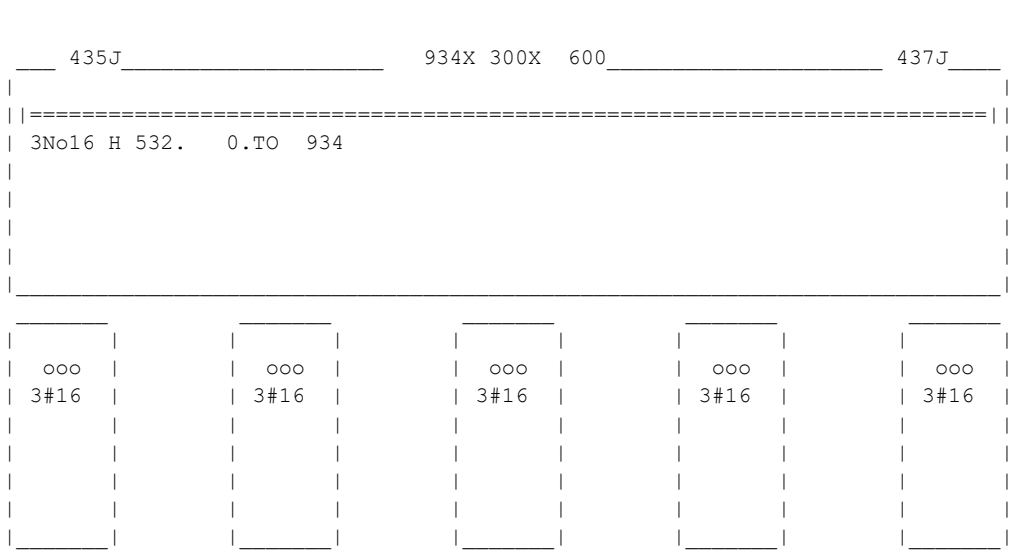
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 1045. MM2, MAX. STEEL PERMISSIBLE = 3924. MM2
MAX POS MOMENT = 197.88 KN-MET, LOADING 302

1	532.	3 - 16MM	0.	934.	YES	YES

CRITICAL NEG MOMENT= 112.93 KN-MET AT 0.MM, LOAD 311
REQD STEEL= 576.MM2, RHO=0.0036, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 82. MMS
REQD. DEVELOPMENT LENGTH = 410. MMS

Cracked Moment of Inertia Iz at above location = 90032.8 cm^4



BEAM NO. 91 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	2000.	YES	YES

CRITICAL POS MOMENT= 7.81 KN-MET AT 2000.MM, LOAD 307
REQD STEEL= 118.MM2, RHO=0.0017, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 6680.0 cm^4

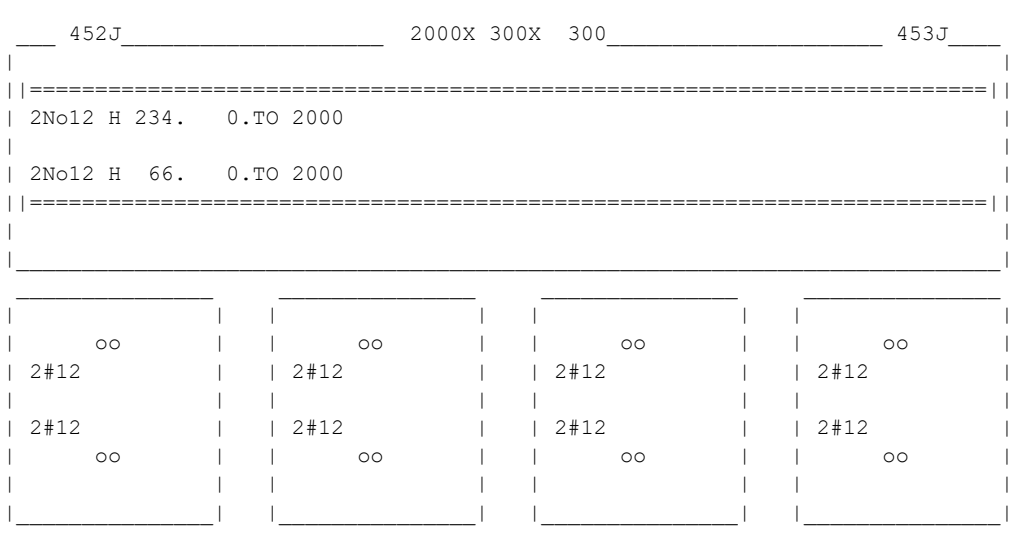
2	234.	2 - 12MM	0.	2000.	YES	YES

CRITICAL NEG MOMENT= 9.74 KN-MET AT 2000.MM, LOAD 302
REQD STEEL= 148.MM2, RHO=0.0021, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 6680.0 cm^4

B E A M N O. 91 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 2.44 KNS Vc= 122.89 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 2.6 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 2.14 KNS Vc= 122.89 KNS Vs= 0.00 KNS
Tu= 0.09 KN-MET Tc= 2.6 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 92 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1000.	YES	YES

CRITICAL POS MOMENT= 1.25 KN-MET AT 1000.MM, LOAD 307
REQD STEEL= 19.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 6680.0 cm^4

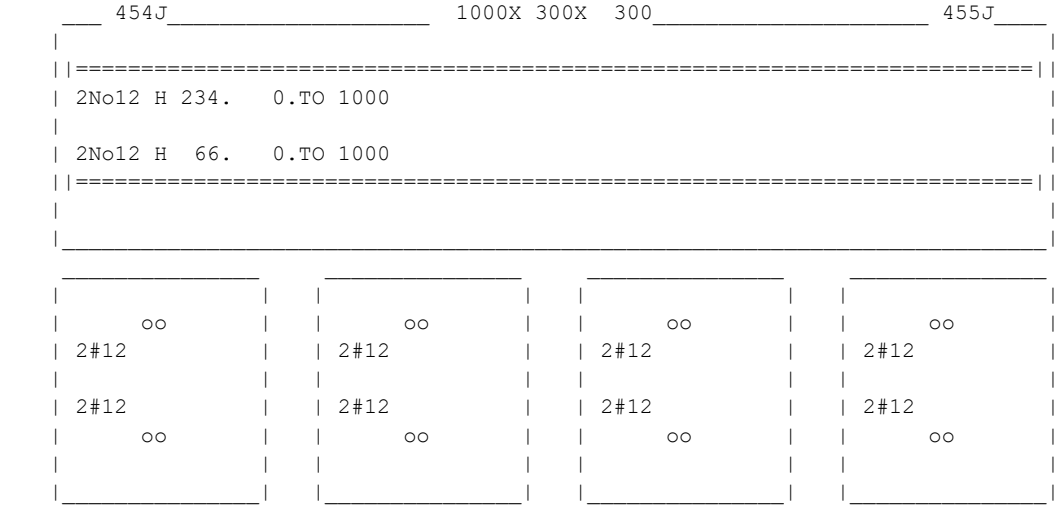
2	234.	2 - 12MM	0.	1000.	YES	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      1.56 KN-MET  AT     0.MM, LOAD  303|
|   REQD STEEL=    24.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =     6680.0 cm^4
```

B E A M N O. 92 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    1.10 KNS  Vc=  121.25 KNS  Vs=    0.00 KNS
 Tu=    0.44 KN-MET  Tc=    2.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    0.50 KNS  Vc=  121.25 KNS  Vs=    0.00 KNS
 Tu=    0.44 KN-MET  Tc=    2.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 93 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1500.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      2.42 KN-MET  AT     0.MM, LOAD  302|
|   REQD STEEL=    37.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =     6680.0 cm^4

  2        234.     2 - 12MM         0.          933.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      2.03 KN-MET  AT     0.MM, LOAD  307|
|   REQD STEEL=    31.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

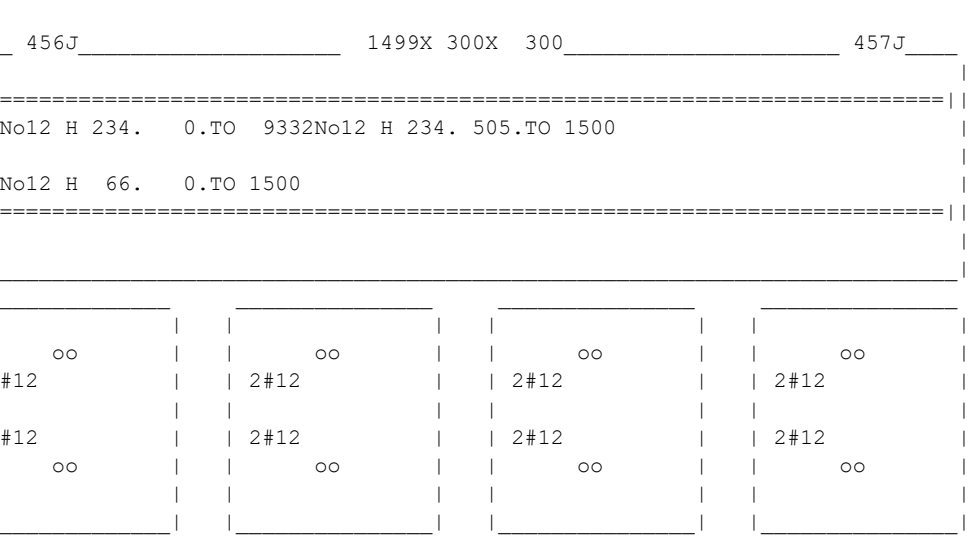
  Cracked Moment of Inertia Iz at above location =     6680.0 cm^4

  3        234.     2 - 12MM       505.         1500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      3.61 KN-MET  AT  1500.MM, LOAD  302|
|   REQD STEEL=    54.MM2, RHO=0.0008, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =     6680.0 cm^4
```

B E A M N O. 93 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    0.28 KNS  Vc=  121.75 KNS  Vs=    0.00 KNS
Tu=    0.27 KN-MET  Tc=    2.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    2.81 KNS  Vc=   67.13 KNS  Vs=    0.00 KNS
Tu=    0.27 KN-MET  Tc=    2.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 94 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

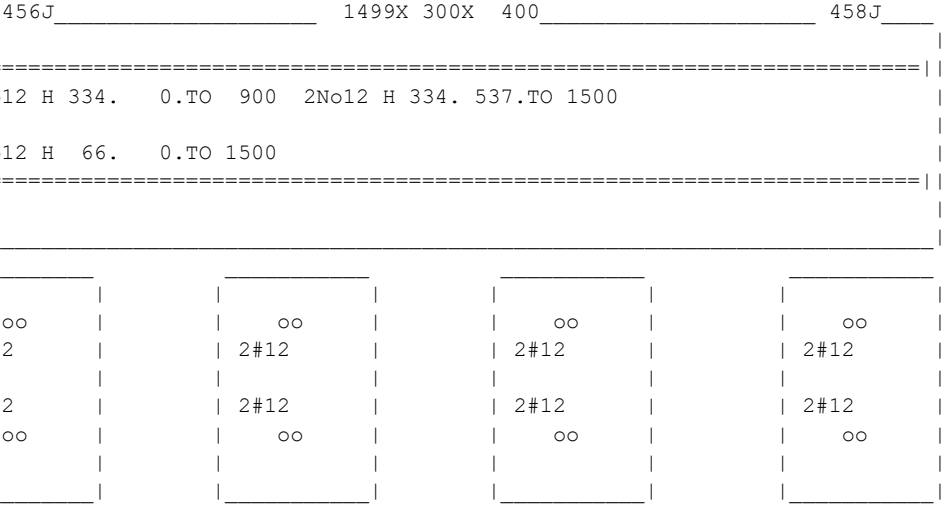
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1500.	YES	YES
	CRITICAL POS MOMENT= 3.53 KN-MET AT 0.MM, LOAD 304					
	REQD STEEL= 37.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 14234.5 cm^4					
2	334.	2 - 12MM	0.	900.	YES	NO
	CRITICAL NEG MOMENT= 1.74 KN-MET AT 0.MM, LOAD 309					
	REQD STEEL= 18.MM2, RHO=0.0002, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 14234.5 cm^4					
3	334.	2 - 12MM	537.	1500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      3.92 KN-MET  AT  1500.MM, LOAD  304|
|   REQD STEEL=    41.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS        |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14234.5 cm^4
```

B E A M N O. 94 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    0.45 KNS  Vc=   97.98 KNS  Vs=    0.00 KNS
 Tu=    0.41 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    3.78 KNS  Vc=   97.98 KNS  Vs=    0.00 KNS
 Tu=    0.41 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 95 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1500.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      1.95 KN-MET  AT  1500.MM, LOAD  305|
|   REQD STEEL=    21.MM2, RHO=0.0002, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

```
 2        334.     2 - 12MM         0.         1025.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      3.41 KN-MET  AT     0.MM, LOAD  305|
|   REQD STEEL=    36.MM2, RHO=0.0004, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|
```

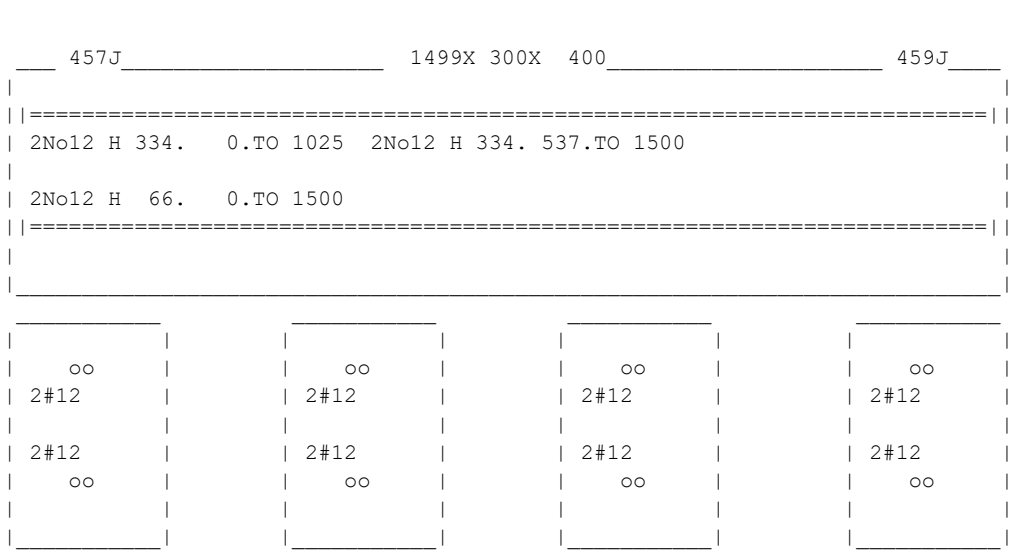
Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

```
 3        334.     2 - 12MM       537.         1500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      1.54 KN-MET  AT  1500.MM, LOAD  308|
|   REQD STEEL=    16.MM2, RHO=0.0002, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 14234.5 cm^4

B E A M N O. 95 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    2.99 KNS  Vc=   96.03 KNS  Vs=    0.00 KNS
Tu=    0.34 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    0.33 KNS  Vc=   93.71 KNS  Vs=    0.00 KNS
Tu=    0.34 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 96 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 300. MMS

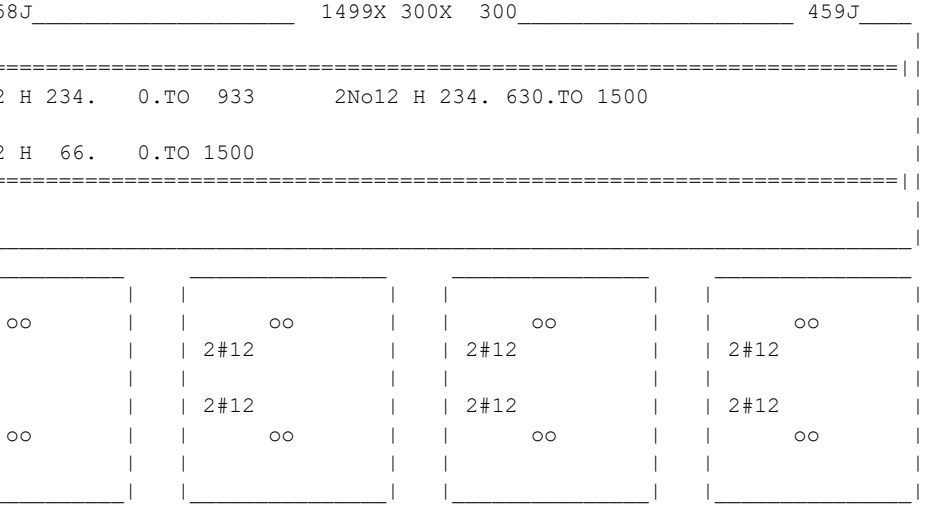
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	2 - 12MM	0.	1500.	YES	YES
	CRITICAL POS MOMENT= 1.38 KN-MET AT 1500.MM, LOAD 303					
	REQD STEEL= 21.MM2, RHO=0.0003, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 6680.0 cm^4					
2	234.	2 - 12MM	0.	933.	YES	NO
	CRITICAL NEG MOMENT= 2.48 KN-MET AT 0.MM, LOAD 303					
	REQD STEEL= 37.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 168. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 6680.0 cm^4					
3	234.	2 - 12MM	630.	1500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      0.87 KN-MET  AT  1500.MM, LOAD  306|
|   REQD STEEL=    13.MM2, RHO=0.0002, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./  168. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =     6680.0 cm^4
```

B E A M N O. 96 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    2.84 KNS  Vc=   66.11 KNS  Vs=    0.00 KNS
 Tu=    0.37 KN-MET  Tc=    2.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                   STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=    0.25 KNS  Vc=  121.47 KNS  Vs=    0.00 KNS
 Tu=    0.37 KN-MET  Tc=    2.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                   STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 97 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5250. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	68.	4 - 16MM	0.	5250.	YES	YES

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=     99.54 KN-MET  AT  5250.MM, LOAD  305|
|   REQD STEEL=   631.MM2, RHO=0.0049, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   55. MMS         |
|   REQD. DEVELOPMENT LENGTH =   601. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    72866.2 cm^4

  2        432.     4 - 16MM         0.         2721.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=    113.52 KN-MET  AT     0.MM, LOAD  305|
|   REQD STEEL=   724.MM2, RHO=0.0056, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   55. MMS         |
|   REQD. DEVELOPMENT LENGTH =   601. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    72866.2 cm^4

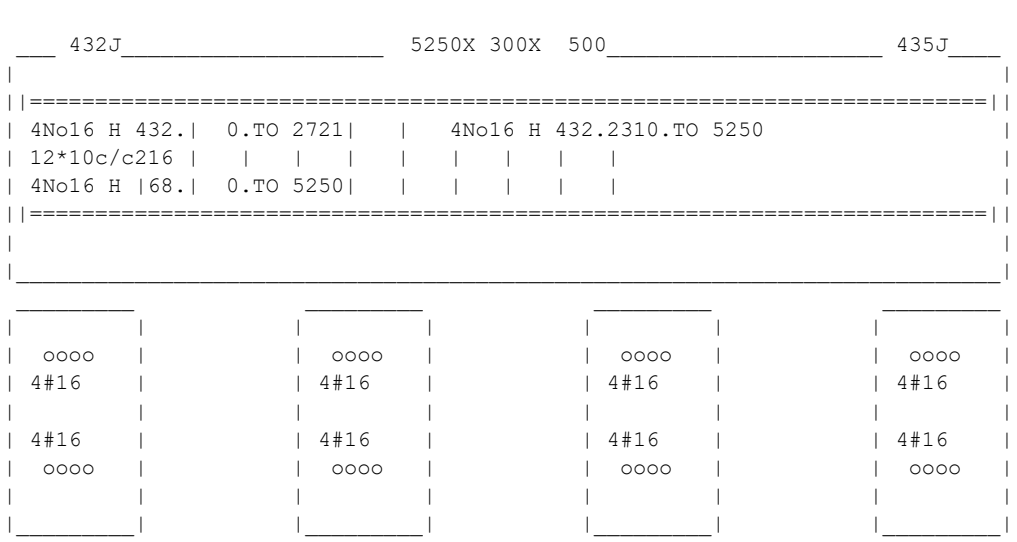
  3        432.     4 - 16MM      2310.         5250.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     95.93 KN-MET  AT  5250.MM, LOAD  308|
|   REQD STEEL=   608.MM2, RHO=0.0047, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   41./   55. MMS         |
|   REQD. DEVELOPMENT LENGTH =   601. MMS                        |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    72866.2 cm^4
```

B E A M N O. 97 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   49.89 KNS  Vc=  123.61 KNS  Vs=    0.00 KNS
Tu=    1.59 KN-MET  Tc=    5.3 KN-MET  Ts=    0.0 KN-MET  LOAD   305
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 216. MM  C/C FOR 2195. MM

AT END   SUPPORT - Vu=    2.44 KNS  Vc=  111.82 KNS  Vs=    0.00 KNS
Tu=    0.98 KN-MET  Tc=    4.6 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 98 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5250. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

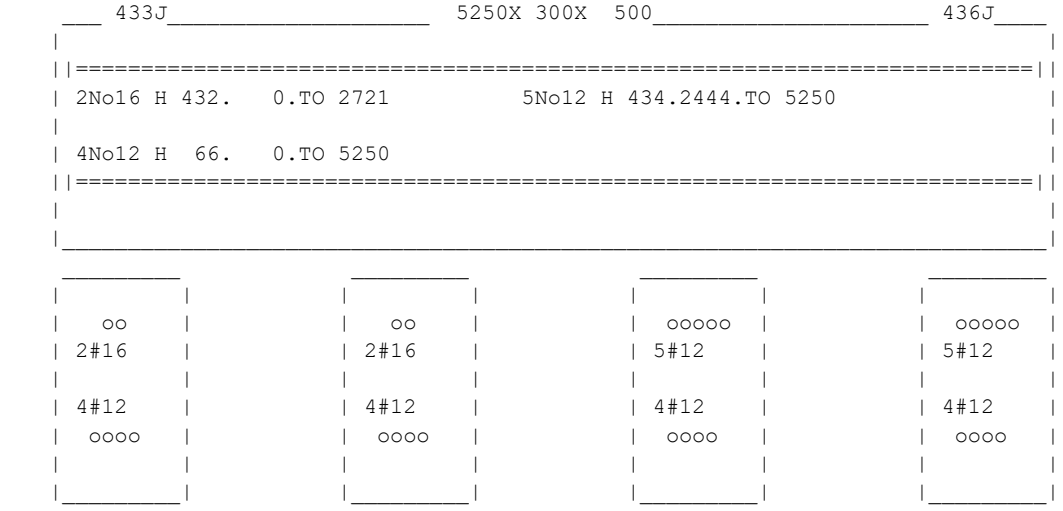
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	0.	5250.	YES	YES
	CRITICAL POS MOMENT= 51.75 KN-MET AT 0.MM, LOAD 304					
	REQD STEEL= 427.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 45474.8 cm^4					
2	432.	2 - 16MM	0.	2721.	YES	NO
	CRITICAL NEG MOMENT= 44.43 KN-MET AT 0.MM, LOAD 309					
	REQD STEEL= 367.MM2, RHO=0.0028, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 164. MMS					
	REQD. DEVELOPMENT LENGTH = 410. MMS					
	Cracked Moment of Inertia Iz at above location = 40697.6 cm^4					
3	434.	5 - 12MM	2444.	5250.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     67.95 KN-MET  AT  5250.MM, LOAD  304|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS        |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    54970.5 cm^4
```

B E A M N O. 98 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    1.13 KNS  Vc=  121.85 KNS  Vs=    0.00 KNS
 Tu=    0.07 KN-MET  Tc=    5.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   20.60 KNS  Vc=  125.07 KNS  Vs=    0.00 KNS
 Tu=    0.07 KN-MET  Tc=    5.5 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 99 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	3000.	YES	YES

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=     63.85 KN-MET  AT     0.MM, LOAD  308|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                        |
|-------------------------------------------------------------|

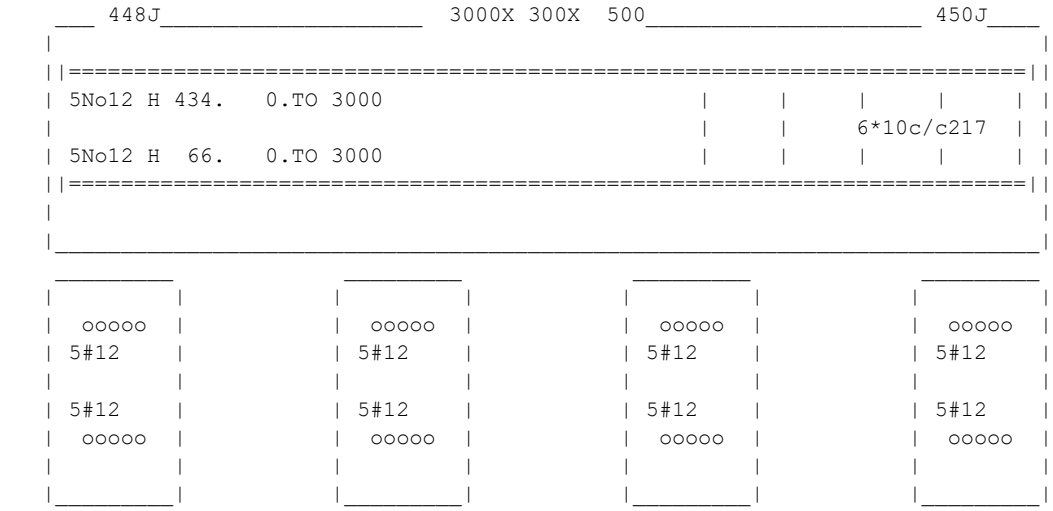
  Cracked Moment of Inertia Iz at above location =    54970.5 cm^4

  2        434.     5 - 12MM         0.         3000.        YES  YES
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     69.40 KN-MET  AT  3000.MM, LOAD  304|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                        |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    54970.5 cm^4
```

B E A M N O. 99 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=    0.70 KNS  Vc=  127.73 KNS  Vs=    0.00 KNS
  Tu=    0.27 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                   STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=   48.60 KNS  Vc=  125.68 KNS  Vs=    0.00 KNS
  Tu=    3.45 KN-MET  Tc=    5.1 KN-MET  Ts=    0.0 KN-MET  LOAD   304
  NO STIRRUPS ARE REQUIRED FOR TORSION.
  REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
  PROVIDE 10 MM 2-LEGGED STIRRUPS AT 217. MM  C/C FOR 1070. MM
```



BEAM NO. 100 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	5 - 12MM	0.	3000.	YES	YES

CRITICAL POS MOMENT= 59.57 KN-MET AT 3000.MM, LOAD 305
REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 54970.5 cm^4

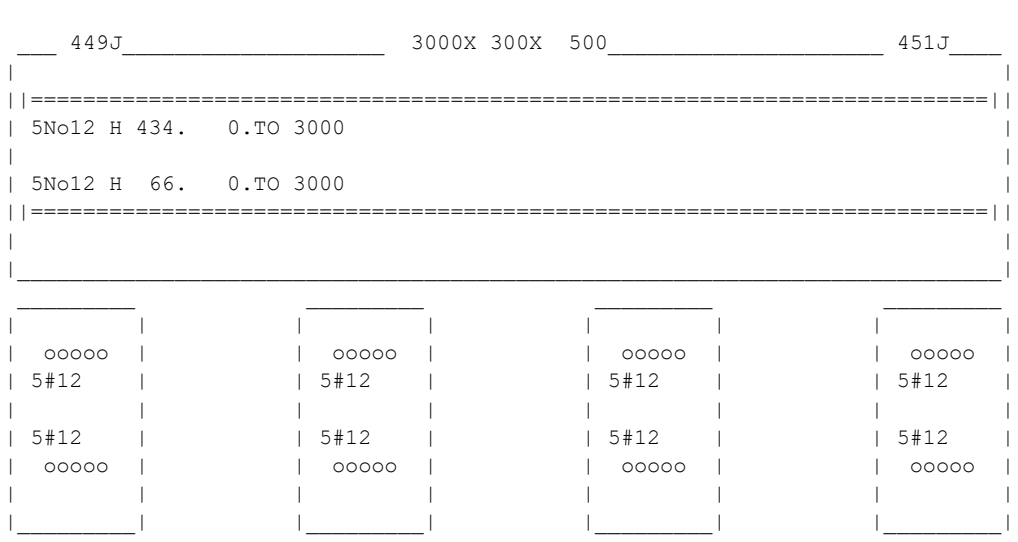
2	434.	5 - 12MM	0.	3000.	YES	YES

CRITICAL NEG MOMENT= 55.94 KN-MET AT 3000.MM, LOAD 308
REQD STEEL= 457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 42. MMS
REQD. DEVELOPMENT LENGTH = 440. MMS

Cracked Moment of Inertia Iz at above location = 54970.5 cm^4

B E A M N O. 100 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 9.91 KNS Vc= 127.27 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 0.68 KNS Vc= 122.91 KNS Vs= 0.00 KNS
Tu= 1.64 KN-MET Tc= 5.2 KN-MET Ts= 0.0 KN-MET LOAD 301
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 101 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END

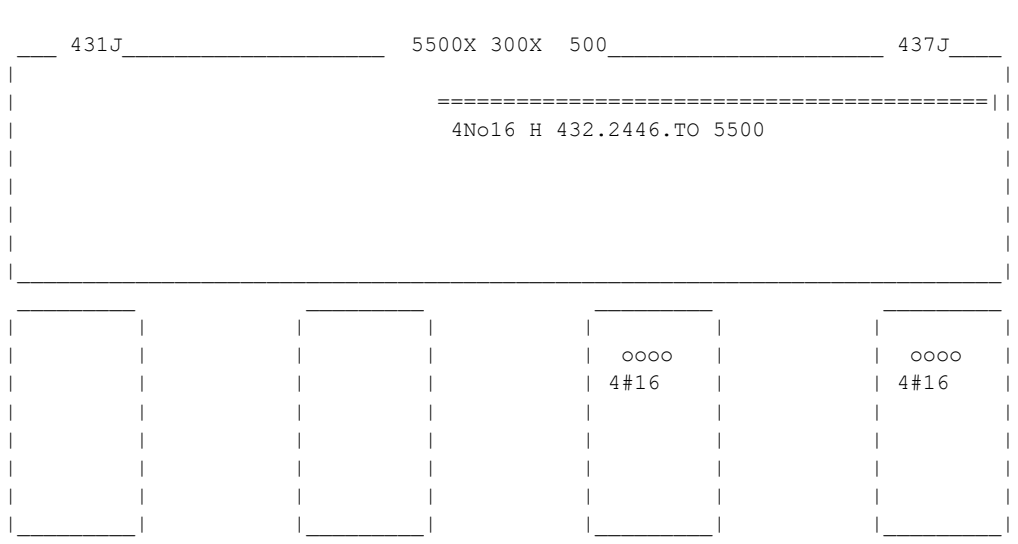
*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 812. MM2, MAX. STEEL PERMISSIBLE = 3177. MM2
MAX POS MOMENT = 124.77 KN-MET, LOADING 305

*** A SUITABLE BAR ARRANGEMENT COULD NOT BE DETERMINED.
REQD. STEEL = 813. MM2, MAX. STEEL PERMISSIBLE = 3177. MM2
MAX NEG MOMENT = 124.99 KN-MET, LOADING 305

1	432.	4 - 16MM	2446.	5500.	NO	YES

CRITICAL NEG MOMENT= 117.22 KN-MET AT 5500.MM, LOAD 308
REQD STEEL= 748.MM2, RHO=0.0058, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 225./ 41./ 55. MMS
REQD. DEVELOPMENT LENGTH = 601. MMS

Cracked Moment of Inertia Iz at above location = 72866.2 cm^4



BEAM NO. 102 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

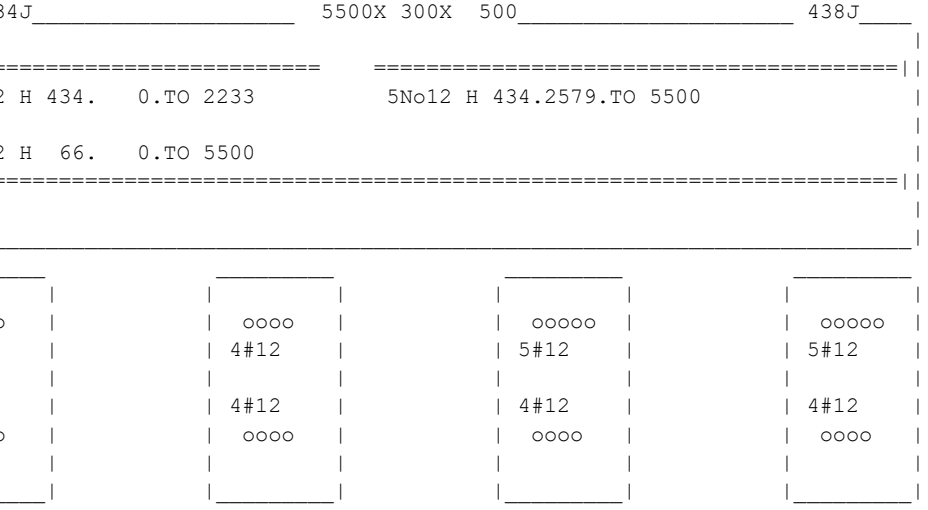
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	66.	4 - 12MM	0.	5500.	YES	YES
	CRITICAL POS MOMENT= 53.59 KN-MET AT 0.MM, LOAD 304					
	REQD STEEL= 442.MM2, RHO=0.0034, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 45474.8 cm^4					
2	434.	4 - 12MM	0.	2233.	YES	NO
	CRITICAL NEG MOMENT= 52.36 KN-MET AT 0.MM, LOAD 309					
	REQD STEEL= 432.MM2, RHO=0.0033, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 225./ 37./ 56. MMS					
	REQD. DEVELOPMENT LENGTH = 330. MMS					
	Cracked Moment of Inertia Iz at above location = 45474.8 cm^4					
3	434.	5 - 12MM	2579.	5500.	NO	YES

```
|---------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     66.51 KN-MET  AT  5500.MM, LOAD  304|
|   REQD STEEL=   457.MM2, RHO=0.0035, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   225./   37./   42. MMS         |
|   REQD. DEVELOPMENT LENGTH =   440. MMS                        |
|---------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    54970.5 cm^4
```

B E A M N O. 102 D E S I G N R E S U L T S - SHEAR

```
  AT START SUPPORT - Vu=    5.93 KNS  Vc=  229.97 KNS  Vs=    0.00 KNS
  Tu=    0.65 KN-MET  Tc=    5.4 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                   STIRRUPS ARE NOT REQUIRED.
  AT END   SUPPORT - Vu=   17.02 KNS  Vc=  125.44 KNS  Vs=    0.00 KNS
  Tu=    0.65 KN-MET  Tc=    5.4 KN-MET  Ts=    0.0 KN-MET  LOAD   301
                   STIRRUPS ARE NOT REQUIRED.
```




```
  ******************END OF BEAM DESIGN*************************


  609. END CONCRETE DESIGN
  610. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= AUG 18,2020 TIME= 12:51:38 ****

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